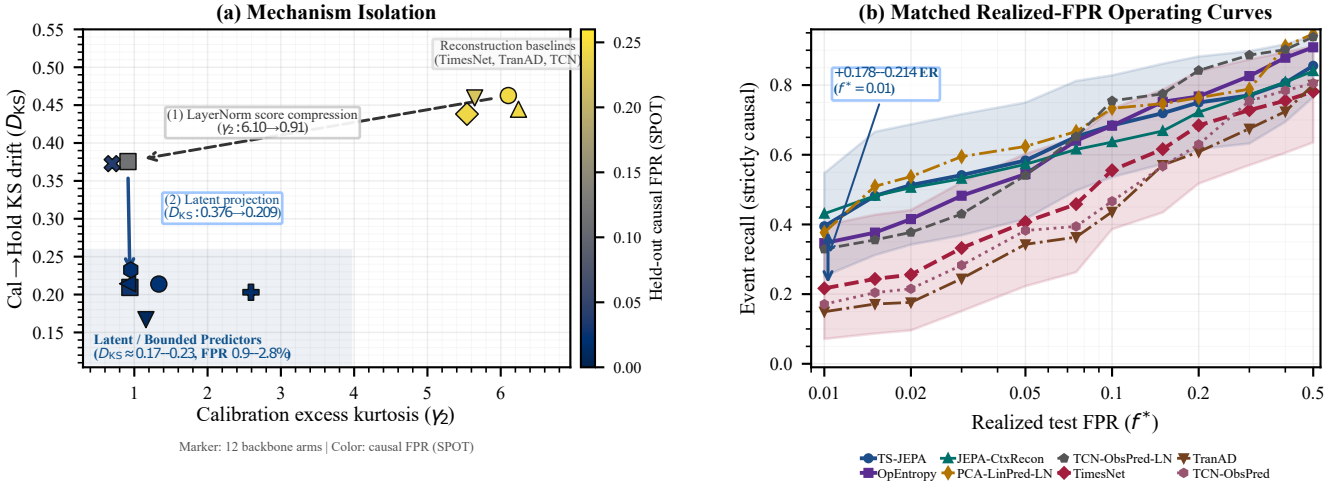

(a) Mechanism Isolation
Reconstruction baselines
(TimesNet, TranAD, TCN)
(1) LayerNorm score compression
(γ_2 : 6.10→0.91)
(2) Latent projection
(D_{KS} : 0.376→0.209)
Latent / Bounded Predictors
(D_{KS} ≈ 0.17–0.23, FPR 0.9–2.8%)
Cal→Hold KS drift (D_{KS})
Calibration excess kurtosis (γ_2)
Held-out causal FPR (SPOT)
Marker: 12 backbone arms | Color: causal FPR (SPOT)
(b) Matched Realized-FPR Operating Curves
+0.178–0.214 ER
(f^* = 0.01)
Event recall (strictly causal)
Realized test FPR (f^*)
TS-JEPA
JEPA-CtxRecon
TCN-ObsPred-LN
TranAD
OpEntropy
PCA-LinPred-LN
TimesNet
TCN-ObsPred